

Q.No :7

Question description

the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly.

Dr.Ramprasad is a faculty, who handling data structure course for EEE department second year students.

one day this faculty was handling very interesting topic in data structure such that Linked List,

During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept.

So the faculty decided to conduct test on the topic of Linked List.

the question was given to last bench students that is,

The nodes are deleted before a certain given node in the linked list.

For example if the given Linked List is 5->10->15->20->25 and

delete before 15 then the Linked List becomes 15->20->25.

Constraint :

$1 < N < 1000$

$1 < P < N-1$

INPUT Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third line contains position of the node to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

Logical Test Cases

Test Case 1

INPUT (STDIN)

4
9 77 12 6
12

EXPECTED OUTPUT

Linked List:->12->6

Test Case 2

INPUT (STDIN)

4
91 77 12 61
2

EXPECTED OUTPUT

Invalid Node! Linked List:->91->77->12->61

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

node *next;

Test Case 3

KEYWORD

void create()

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node* create(int data)
{
    struct node *newNode;
```

```

newNode = (struct node *)malloc(sizeof(struct node));

newNode->data = data;
newNode->next = NULL;

return newNode;
}
void display(struct node *head)
{
    struct node *temp = head;

    printf("Linked List:->");

    while (temp != NULL)
    {
        printf("%d", temp->data);

        if (temp->next != NULL)
            printf("->");

        temp = temp->next;
    }

    printf("\n");
}
struct node* deleteBefore(struct node *head, int p, int *found)
{
    struct node *temp;
    struct node *deleteNode;

    temp = head;
    while (temp != NULL)
    {
        if (temp->data == p)
        {
            *found = 1;
            break;
        }

        temp = temp->next;
    }
    if (*found == 0)
    {
        return head;
    }
    while (head != NULL && head->data != p)
    {
        deleteNode = head;
        head = head->next;
        free(deleteNode);
    }

    return head;
}

```

```

}

int main()
{
int n;
int data;
int p;
int found = 0;

struct node *head = NULL;
struct node *temp;
struct node *newNode;

printf("Input value:\n");

scanf("%d", &n);
for (int i = 0; i < n; i++)
{
scanf("%d", &data);

newNode = create(data);

if (head == NULL)
{
head = newNode;
}
else
{
temp = head;

while (temp->next != NULL)
{
temp = temp->next;
}

temp->next = newNode;
}
scanf("%d", &p);
head = deleteBefore(head, p, &found);

if (found == 0)
{
printf("Invalid Node! ");
}
display(head);

return 0;
}

```

Output:

Test_case 1:

Input value:

4

9 77 12 6

12

Linked List:->12->6

Test_case 2:

Input value:

4

91 77 12 61

2

Invalid Node! Linked List:->91->77->12->61